Gregory Polyakov

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Summary

Third-year PhD student at the University of Tübingen and IMPRS-IS working on mechanistic interpretability of language models for NLP and IR. Studies how models implement complex behaviors such as hallucination, in-context learning, and relevance estimation, and turns these findings into methods for controlling and improving them. First author of four papers, including at EMNLP and SIGIR, with three years of industrial R&D at Huawei Noah's Ark Lab on production search and voice-assistant systems.

Education

University of Tübingen & Max Planck Research School — *Tübingen, Germany*
***PhD** in Computer Science* — *Oct 2024 – Aug 2028 (Expected)*

- Health-NLP Lab, Supervisor: Prof. Dr. Carsten Eickhoff.
- Scholar, International Max Planck Research School for Intelligent Systems (IMPRS-IS).
- Research focus: Mechanistic interpretability of language models for NLP and IR tasks.
- Available for an internship in 2027 (flexible dates and duration).

Moscow Institute of Physics and Technology — *Moscow, Russia*
***MSc** in Applied Mathematics and Computer Science* — *Sep 2021 – Jul 2024*

- Research topics: sparse retrieval, multimodal retrieval, tool calling in LLMs.
- Thesis: "Approaches to Training Language Models for External Function Calling: Using Self-Generated Data for Better Error Detection and Recovery".

***BSc** in Applied Mathematics and Computer Science* — *Sep 2017 – Jul 2021*

- Thesis: "Usage of MCMC Methods in Deep Learning".

Publications

Predicting Hallucinations Before, During, and After Generation with Sparse Autoencoder Features — *Under review*
***G. Polyakov**, A. Bahrainian, C. Eickhoff*

- Systematically studied hallucination detection from model internals before, during, and after generation, showing that a substantial signal exists before generation.
- Proposed PreSAGE, a trainable SAE-based detector with gated feature selection and attention pooling, best in 19 of 24 QA settings and the most consistent overall.

Understanding Wacky Weights: A Dissection of SPLADE's Learned Term Importance — *SIGIR 2026 (Reproducibility)*
***G. Polyakov**, H. Scells, C. Eickhoff*

- Introduced the Wackiness Score and W-AUC to quantify semantically unrelated expansion terms across models with different vocabularies and sparsity levels.
- Trained 6 SPLADE variants, showing that the prevalence of wacky tokens increases with vocabulary size and that they mainly aid in-domain retrieval.

Interpretability Analysis of Arithmetic In-Context Learning in Large Language Models — *EMNLP 2025 (Main)*
***G. Polyakov**, C. Hepting, C. Eickhoff, A. Bahrainian*

- Traced how 4 LLMs (up to 12B) use in-context examples for multi-step arithmetic, showing that format consistency matters more than arithmetic correctness.
- Distilled in-context examples into a function vector that raises zero-shot accuracy by up to 25 points and fully recovers one-shot accuracy on Pythia-12B.

Towards Best Practices of Axiomatic Activation Patching in Information Retrieval — *SIGIR 2025 (Short)*
***G. Polyakov**, C. Chen, C. Eickhoff*

- Proposed a noise-mitigating evaluation metric for activation patching in neural rankers that filters out spurious low-confidence signals.
- Systematically analyzed diagnostic datasets, identifying key selection criteria for queries and documents to ensure robust interpretability analysis.

ToolReflection: Improving Large Language Models for Real-World API Calls with Self-Generated Data *ACL 2025 (REALM Workshop)*

***G. Polyakov**, I. Alimova, D. Abulkhanov, I. Sedykh, A. Bout, S. Nikolenko, I. Piontkovskaya*

- Proposed a fine-tuning strategy that enables LLMs to self-correct API calls by learning from runtime execution feedback.
- Co-developed evaluation datasets for tool-augmented LLMs in realistic scenarios.

Sinkhorn Transformations for Single-Query Postprocessing in Text-Video Retrieval *SIGIR 2023 (Short)*

*K. Yakovlev, **G. Polyakov**, I. Alimova, A. V. Podolskiy, A. Bout, S. I. Nikolenko, I. Piontkovskaya*

Work Experience

Huawei Noah's Ark Lab *Moscow, Russia*

NLP Research Engineer, full-time *Nov 2022 – Sep 2024*

- Developed a fine-tuning method that teaches LLMs to detect and recover from failed API calls (ACL 2025 REALM Workshop).
- Designed and co-implemented the end-to-end NLU pipeline for a production in-car voice assistant in Russian.
- Co-developed a Sinkhorn-based post-processing method that improves text-video retrieval quality (SIGIR 2023).

NLP Research Intern, full-time *Oct 2021 – Nov 2022*

- Co-developed the reranking model for a large-scale Russian-language product search engine: improved ranking quality by 5% relative NDCG@10 and cut inference latency 1.5x by adapting ColBERTer and TinyBERT-style distillation.

Invited Talks

Understanding Wacky Weights: A Dissection of SPLADE's Learned Term Importance *Oct 2026*

Glasgow IR Seminar, University of Glasgow

Funding & Scholarships

Presidential Grant Scholarship *Feb 2018 – Aug 2021*

Four-year stipend for academic and competition merit, one of 1,200 recipients nationwide.

Teaching Experience

University of Tübingen

Teaching Assistant, Introduction to AI *Winter Semester 2026/27*

- Conducted hands-on tutorials and evaluated student assignments.

Teaching Assistant, Modern Search Engines *Summer Semesters 2025, 2026*

- Conducted hands-on tutorials, evaluated assignments, and mentored student projects.

Teaching Assistant, Introduction to Python Programming *Winter Semesters 2024/25, 2025/26*

- Conducted hands-on tutorials and evaluated student assignments.

Skills

Programming: Python, C/C++, Bash.
ML & NLP: PyTorch, Hugging Face Transformers, PEFT, LLM fine-tuning (SFT, LoRA), tool use, Scikit-learn.
Interpretability: TransformerLens, activation patching, steering, probing, SAEs, cross-layer transcoders.
IR: Pyserini, FAISS, sparse and dense retrieval, neural reranking, knowledge distillation.
Infrastructure: SLURM, multi-GPU training (HF Accelerate), Git.
Languages: Russian (native), English (C1, IELTS Band Score 7).